The Gaussian Correlation Inequality

Slide edition

After T. Royen (2014) and R. Latała & D. Matlak (2015)

1 Introduction

Theorem 1.1 (Gaussian correlation inequality)**.** For any closed symmetric convex sets $K, L \subseteq \mathbb{R}^d$ and any centered Gaussian measure μ on $\mathbb{R}^d$,

$$\mu(K \cap L) \geqslant \mu(K)\,\mu(L). \tag{1}$$

These slides prove Theorem 1.1 by Royen's analytic argument, following Latała–Matlak's Gaussian-only exposition. The inputs are the Laplace transform of a squared Gaussian, a determinant expansion, dominated convergence, and integration by parts.

Notation 1.2.

- $\mathcal{N}(0, C)$: centered Gaussian distribution with covariance C.
- $M_{n\times m}$: real $n \times m$ matrices. $A \succ 0$ (resp. $A \succcurlyeq 0$) means A is symmetric positive (semi)definite.
- For $A = (a_{ij})_{i,j\leqslant n}$ and $J \subseteq [n] := \{1, \dots, n\}$, $A_J := (a_{ij})_{i,j\in J}$ is the principal submatrix, $|A|$ the determinant, $|J|$ the cardinality.
- I_n: $n \times n$ identity.
- For $\varepsilon \in \{-1, 1\}^n$, $x \in (0, \infty)^n$: $\sqrt{x}_\varepsilon := (\varepsilon_i \sqrt{x_i})_{i\leqslant n}$.

We will use three standard facts without further comment.

Lemma 1.3 (Square root)**.** For any $A \succcurlyeq 0$ there exists a unique $A^{1/2} \succcurlyeq 0$ with $(A^{1/2})^2 = A$, and $A^{1/2}$ is invertible iff $A \succ 0$.

(Spectral theorem: diagonalize $A = O\Lambda O^\top$ with $\Lambda \succcurlyeq 0$, set $A^{1/2} := O\Lambda^{1/2}O^\top$; uniqueness is standard.)

Lemma 1.4 (Sylvester determinant identity)**.** For $A_1 \in M_{n\times m}$ and $A_2 \in M_{m\times n}$, $|I_n + A_1A_2| = |I_m + A_2A_1|$.

Proof. Compute $D := \left|\begin{smallmatrix} I_n & -A_1 \\ A_2 & I_m \end{smallmatrix}\right|$ two ways. Row-reduce $R_1 \leftarrow R_1 + A_1R_2$ to clear the $(1,1)$-block's contribution to the $(1,2)$ block: the matrix becomes $\left(\begin{smallmatrix} I_n+A_1A_2 & 0 \\ A_2 & I_m \end{smallmatrix}\right)$, so $D = |I_n + A_1A_2|$. Alternatively, column-reduce $C_1 \leftarrow C_1 + C_2A_2$: the matrix becomes $\left(\begin{smallmatrix} I_n & -A_1 \\ 0 & I_m+A_2A_1 \end{smallmatrix}\right)$, so $D = |I_m + A_2A_1|$. $\square$

Lemma 1.5 (Strict separation; Hahn–Banach)**.** Let $K_1, K_2 \subseteq \mathbb{R}^d$ be disjoint closed convex sets with K_1 bounded. There exist $w \in \mathbb{R}^d \setminus \{0\}$ and $\alpha_1 < \alpha_2$ in $\mathbb{R}$ with $\langle w, x\rangle \leqslant \alpha_1$ for $x \in K_1$ and $\langle w, x\rangle \geqslant \alpha_2$ for $x \in K_2$.

(Standard corollary of Hahn–Banach in finite dimension; see e.g. Rockafellar, *Convex Analysis*, Cor. 11.4.2.)

2 Setup and reduction to rectangles

Definition 2.1 (Centered Gaussian measure)**.** For $C \succ 0$, the measure $\mathcal{N}(0, C)$ on $\mathbb{R}^n$ has density

$$\phi_C(x) \;=\; (2\pi)^{-n/2}|C|^{-1/2}\exp\bigl(-\tfrac{1}{2}\langle C^{-1}x, x\rangle\bigr).$$

We assume $C \succ 0$ throughout the proof.

Definition 2.2. $K \subseteq \mathbb{R}^n$ is *symmetric* if $K = -K$. A *symmetric slab* is $S_{v,t} := \{x \in \mathbb{R}^n : |\langle v, x\rangle| \leqslant t\}$ for $v \in \mathbb{R}^n \setminus \{0\}$, $t > 0$.

Lemma 2.3 (Slab approximation)**.** Every closed symmetric convex set $K \subseteq \mathbb{R}^d$ is a countable intersection of symmetric slabs $S_{v,t}$.

Proof.

1. Reduction to bounded K. Set $K_m := K \cap [-m, m]^d$, a closed symmetric convex set. Since $K = \bigcup_m K_m$, by monotone convergence $\mu(K) = \lim_m \mu(K_m)$ and similarly for $K \cap L$, so it suffices to prove the slab approximation for bounded K; we assume $K \subseteq B(0, R)$ for some $R < \infty$ below.
2. Bounded case. Let $\mathcal{V} := \{(v, t) \in \mathbb{Q}^d \times \mathbb{Q}_{>0} : K \subseteq S_{v,t}\}$, countable. Trivially $K \subseteq \bigcap_{(v,t)\in\mathcal{V}} S_{v,t}$.

   Conversely, fix $y \notin K$ and let $\varepsilon > 0$ with $B(y, \varepsilon) \cap K = \varnothing$. Applying Theorem 1.5 to $K_1 := \overline{B(y, \varepsilon/2)}$ (bounded) and $K_2 := K$ (closed convex, disjoint from K_1) yields $w \in \mathbb{R}^d \setminus \{0\}$ and $a \in \mathbb{R}$ with $\sup_{x\in K}\langle w, x\rangle \leqslant a$ and $\langle w, y\rangle \geqslant a + \varepsilon|w|$. Since K is symmetric and convex, $0 = \frac{1}{2}x + \frac{1}{2}(-x) \in K$ for any $x \in K$, hence $a \geqslant \langle w, 0\rangle = 0$; and $K = -K$ gives $\sup_{x\in K} |\langle w, x\rangle| \leqslant a$. Pick $(w', a') \in \mathbb{Q}^d \times \mathbb{Q}_{>0}$ with $|w' - w| < \delta$, $|a' - a| < \delta$. Then

$$\sup_{x\in K} |\langle w', x\rangle| \leqslant a + \delta R, \qquad |\langle w', y\rangle| \geqslant a + \varepsilon|w| - \delta|y|.$$

   Choosing $\delta < \min(\varepsilon|w|/(|y| + R + 1), \varepsilon|w|/2)$ and $a' := a + \delta R + \delta/2 \in \mathbb{Q}_{>0}$ gives $K \subseteq S_{w',a'}$ and $y \notin S_{w',a'}$. □

By Theorem 2.3 and monotone convergence, the GCI reduces to the case where K and L are finite intersections of

symmetric slabs. Writing

$$K = \{x : |\langle v_i, x\rangle| \leqslant t_i,\ 1 \leqslant i \leqslant n_1\}, \qquad L = \{x : |\langle v_i, x\rangle| \leqslant t_i,\ n_1{+}1 \leqslant i \leqslant n\},$$

setting $n = n_1 + n_2$ and $X_i := \langle v_i, G\rangle$ for $G \sim \mu$, the GCI reduces to:

Theorem 2.4 (GCI, analytic form)**.** Let $X = (X_1, \dots, X_n)$ be centered Gaussian with $C \succ 0$, $n = n_1 + n_2$. For any $t_1, \dots, t_n > 0$,

$$\mathsf{P}(|X_i| \leqslant t_i,\ i \leqslant n) \ \geqslant\ \mathsf{P}(|X_i| \leqslant t_i,\ i \leqslant n_1)\,\mathsf{P}(|X_i| \leqslant t_i,\ n_1{<}i \leqslant n). \tag{2}$$

3 Auxiliary lemmas

3.1 Laplace transform of squared Gaussian vectors

Lemma 3.1. For $X \sim \mathcal{N}(0, C)$ in $\mathbb{R}^n$ and $\lambda_i \geqslant 0$,

$$\mathsf{E}\exp\Big(-\sum_{i=1}^{n} \lambda_i X_i^2\Big) \ =\ |I_n + 2\Lambda C|^{-1/2}, \qquad \Lambda := \operatorname{diag}(\lambda_1, \dots, \lambda_n).$$

Proof.

1. Standard normal quadratic. For $Y \sim \mathcal{N}(0, I_n)$ and a symmetric matrix B with $I_n - 2B \succ 0$, combine the

density of Y with the factor $e^{\langle BY,Y\rangle}$: the exponent is $-\frac{1}{2}\langle y,y\rangle + \langle By,y\rangle = -\frac{1}{2}\langle (I_n - 2B)y,y\rangle$, so

$$\mathsf{E}e^{\langle BY,Y\rangle} = (2\pi)^{-n/2}\int_{\mathbb{R}^n} e^{-\frac{1}{2}\langle (I_n-2B)y,y\rangle}\,\mathrm{d}y = |I_n - 2B|^{-1/2}, \tag{3}$$

using $\int_{\mathbb{R}^n} e^{-\frac{1}{2}\langle Ay,y\rangle}\mathrm{d}y = (2\pi)^{n/2}|A|^{-1/2}$ (change variables $y = A^{-1/2}z$; $A = I_n - 2B$).

2. Reduction to the standard case. Set $A := C^{1/2}$ (Theorem 1.3); then $C = AA^\top$ and $X := AY \sim \mathcal{N}(0,C)$. Taking $B := -A^\top\Lambda A$ in (3): since $\lambda_i \geqslant 0$ gives $\Lambda \succcurlyeq 0$, hence $B \preccurlyeq 0$ and $I_n - 2B \succcurlyeq I_n \succ 0$. The exponent becomes $\langle BY,Y\rangle = -\langle \Lambda AY, AY\rangle = -\sum_i \lambda_i (AY)_i^2 = -\sum_i \lambda_i X_i^2$, so

$$\mathsf{E}\exp\Big(-\sum_i \lambda_i X_i^2\Big) = |I_n + 2A^\top\Lambda A|^{-1/2}.$$

3. Commuting the factors. Apply Theorem 1.4 with $A_1 := A^\top$, $A_2 := 2\Lambda A$: $|I_n + 2A^\top\Lambda A| = |I_n + 2\Lambda AA^\top| = |I_n + 2\Lambda C|$.

□

3.2 A determinant identity

Lemma 3.2.

(i) For $A \in M_{n\times n}$,

$$|I_n + A| \;=\; 1 + \sum_{\varnothing \neq J \subseteq [n]} |A_J|. \tag{4}$$

(ii) For $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{smallmatrix}\right)$ with A_{ii} invertible,

$$|A| = |A_{11}||A_{22}| \cdot \left|I_{n_1} - A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2}\right|. \tag{5}$$

If $A \succ 0$, then $0 \leqslant A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2} \leqslant I_{n_1}$.

Proof.

(i) By Leibniz, $|I_n + A| = \sum_\sigma \operatorname{sgn}(\sigma) \prod_i (\delta_{i\sigma(i)} + a_{i\sigma(i)})$. Expanding the product, each summand corresponds to a choice of $J \subseteq [n]$ (positions where $a_{i\sigma(i)}$ is selected); non-zero contributions require $\sigma(i) = i$ for $i \notin J$. The σ's fixing $[n] \setminus J$ correspond to permutations of J, giving $\sum_J |A_J|$; the $J = \varnothing$ term is 1.

(ii) Let $A_{11}^{1/2}, A_{22}^{1/2}$ be the positive symmetric square roots (Theorem 1.3), invertible by assumption. Setting $P := A_{11}^{-1/2} A_{12} A_{22}^{-1/2}$ and $Q := A_{22}^{-1/2} A_{21} A_{11}^{-1/2}$, a direct block multiplication gives

$$A = \operatorname{diag}(A_{11}^{1/2}, A_{22}^{1/2}) \left(\begin{smallmatrix} I & P \\ Q & I \end{smallmatrix}\right) \operatorname{diag}(A_{11}^{1/2}, A_{22}^{1/2}),$$

so $|A| = |A_{11}||A_{22}| \cdot \left|\begin{smallmatrix} I & P \\ Q & I \end{smallmatrix}\right|$. Row-reduce by $R_1 \leftarrow R_1 - PR_2$ (determinant invariant):

$$\left(\begin{smallmatrix} I & P \\ Q & I \end{smallmatrix}\right) \to \left(\begin{smallmatrix} I-PQ & 0 \\ Q & I \end{smallmatrix}\right),$$

block-triangular with determinant $|I - PQ|$. Substituting $PQ = A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2}$ gives (5).
When $A \succ 0$: for all $s, t \in \mathbb{R}$ and $x \in \mathbb{R}^{n_1}$, $y \in \mathbb{R}^{n_2}$,

$$\langle A(sx, ty)^\top, (sx, ty)^\top \rangle = s^2 \langle A_{11}x, x\rangle + 2st\langle A_{21}x, y\rangle + t^2\langle A_{22}y, y\rangle \geqslant 0.$$

Nonnegativity of this quadratic in (s, t) forces $\langle A_{21}x, y\rangle^2 \leqslant \langle A_{11}x, x\rangle\langle A_{22}y, y\rangle$. Setting $x := A_{11}^{-1/2}u$, $y := A_{22}^{-1/2}v$: $\langle Qu, v\rangle^2 \leqslant |u|^2|v|^2$, i.e., $\|Q\|_{\mathrm{op}} \leqslant 1$, equivalently $0 \preccurlyeq Q^\top Q \preccurlyeq I_{n_1}$. For A symmetric, $Q = P^\top$, so $A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2} = PP^\top = Q^\top Q$.

□

3.3 Density of $Z(\tau)$ and the interchange lemma

Partition

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}, \quad C_{ij} \in M_{n_i \times n_j},\ n_1 + n_2 = n, \tag{6}$$

and define the interpolating family

$$C(\tau) := \begin{pmatrix} C_{11} & \tau C_{12} \\ \tau C_{21} & C_{22} \end{pmatrix} = \tau C(1) + (1-\tau)C(0), \qquad \tau \in [0, 1]. \tag{7}$$

Since $C(0) = \mathrm{diag}(C_{11}, C_{22}) \succ 0$ and $C(1) = C \succ 0$, each $C(\tau) \succ 0$ by convexity. Let $X(\tau) \sim \mathcal{N}(0, C(\tau))$ and $Z_i(\tau) := \frac{1}{2} X_i(\tau)^2$.

Lemma 3.3 (Density and interchange)**.** Let $f(x, \tau)$ denote the density of $Z(\tau)$ on $(0, \infty)^n$.

(i)

$$f(x, \tau) = |C(\tau)|^{-1/2} (4\pi)^{-n/2} \frac{1}{\sqrt{x_1 \cdots x_n}} \sum_{\varepsilon \in \{-1,1\}^n} e^{-\langle C(\tau)^{-1} \sqrt{x}_\varepsilon, \sqrt{x}_\varepsilon \rangle} \mathbf{1}_{(0,\infty)^n}(x). \tag{8}$$

(ii) For any Borel $K \subseteq [0, \infty)^n$ and $\lambda_i \geqslant 0$,

$$\int_K e^{-\sum_i \lambda_i x_i} \frac{\partial}{\partial \tau} f(x, \tau) \, \mathrm{d}x \;=\; \frac{\partial}{\partial \tau} \int_K e^{-\sum_i \lambda_i x_i} f(x, \tau) \, \mathrm{d}x. \tag{9}$$

Proof.

(i) The density of $X(\tau)$ is $\phi_{C(\tau)}$ from Theorem 2.1. On each sign orthant $\mathbb{R}^n_\varepsilon := \{y : \mathrm{sgn}(y_i) = \varepsilon_i\}$, the map $y \mapsto x_i := \frac{1}{2} y_i^2$ is a diffeomorphism onto $(0, \infty)^n$ with inverse $y_i = \varepsilon_i \sqrt{2x_i}$ and Jacobian $\mathrm{d}y_i / \mathrm{d}x_i = \varepsilon_i / \sqrt{2x_i}$; absolute Jacobian $\prod_i (2x_i)^{-1/2}$. Summing over the 2^n sign patterns and using $\langle C(\tau)^{-1} y, y \rangle =$

$2\langle C(\tau)^{-1}\sqrt{x}_\varepsilon, \sqrt{x}_\varepsilon\rangle$ (by $y_i = \varepsilon_i\sqrt{2x_i}$):

$$\begin{aligned} f(x,\tau) &= \sum_\varepsilon \phi_{C(\tau)}(\sqrt{2x}_\varepsilon)\cdot\prod_i (2x_i)^{-1/2} \\ &= |C(\tau)|^{-1/2}(2\pi)^{-n/2}2^{-n/2}(x_1\cdots x_n)^{-1/2}\sum_\varepsilon e^{-\langle C(\tau)^{-1}\sqrt{x}_\varepsilon,\sqrt{x}_\varepsilon\rangle}, \end{aligned}$$

which is (8) after combining $(2\pi)^{-n/2}2^{-n/2} = (4\pi)^{-n/2}$.

(ii) We build a τ-uniform integrable dominating function for $\partial_\tau f$ via four steps.

(a) Eigenvalue lower bound. Each $C(\tau) \succ 0$, and $\tau \mapsto \lambda_{\min}(C(\tau)^{-1}) = 1/\lambda_{\max}(C(\tau))$ is continuous and positive on compact $[0,1]$, so bounded below by some $a > 0$. Then $\langle C(\tau)^{-1}\sqrt{x}_\varepsilon, \sqrt{x}_\varepsilon\rangle \geqslant a|\sqrt{x}_\varepsilon|^2 = a\sum_i x_i$.

(b) Derivative formula. Differentiating $C(\tau)C(\tau)^{-1} = I_n$, $\frac{\mathrm{d}}{\mathrm{d}\tau}C(\tau)^{-1} = -C(\tau)^{-1}C'(\tau)C(\tau)^{-1}$, with $C'(\tau) = C(1) - C(0)$ constant. Since $\tau \mapsto C(\tau)^{-1}$ is continuous on compact $[0,1]$, both $|C(\tau)|^{-1/2}$ and $\|\frac{\mathrm{d}}{\mathrm{d}\tau}C(\tau)^{-1}\|_{\mathrm{op}}$ are bounded, say by $M, b < \infty$. Hence $|\partial_\tau|C(\tau)|^{-1/2}| \leqslant M$ and $|\langle\frac{\mathrm{d}}{\mathrm{d}\tau}C(\tau)^{-1}\sqrt{x}_\varepsilon, \sqrt{x}_\varepsilon\rangle| \leqslant b|\sqrt{x}_\varepsilon|^2 = b\sum_i x_i$.

(c) Dominating function. Differentiating (8) in τ (product rule on the prefactor $|C(\tau)|^{-1/2}$ and the exponent), each of the 2^n sign-orthant summands is bounded by

$$\big(M + b\textstyle\sum_i x_i\big)(4\pi)^{-n/2}(x_1\cdots x_n)^{-1/2}e^{-a\sum_i x_i}.$$

Summing over ε and absorbing constants,

$$\sup_{\tau\in[0,1]} |\partial_\tau f(x,\tau)| \leqslant g(x) := c\,\frac{1+\sum_i x_i}{\sqrt{x_1\cdots x_n}} e^{-a\sum_i x_i}$$

for some $c = c(C,n)$. Fubini: $\int_0^\infty x^{-1/2}(1+x)e^{-ax}\,\mathrm{d}x < \infty$, so factoring in each coordinate gives $\int g < \infty$.

(d) Interchange. For any Borel $K \subseteq [0,\infty)^n$ and $\lambda_i \geqslant 0$, $e^{-\sum_i \lambda_i x_i}|\partial_\tau f| \leqslant g$ uniformly in τ. Dominated convergence applied to the difference quotient $(f(x,\tau+h)-f(x,\tau))/h \to \partial_\tau f(x,\tau)$ gives (9).

□

3.4 The noncentral gamma and the h-family

Definition 3.4. For $\alpha > 0$, $y \geqslant 0$, the *noncentral gamma density* is

$$g_\alpha(x,y) := e^{-x-y}\sum_{k=0}^{\infty} \frac{x^{k+\alpha-1}}{\Gamma(k+\alpha)}\frac{y^k}{k!}, \qquad x > 0.$$

Remark 3.5. Equivalently,

$$g_\alpha(x,y) = \sum_{k\geqslant 0} e^{-y}\frac{y^k}{k!} g_{\alpha+k}(x),$$

a Poisson(y) mixture of central gamma($\alpha + k$) densities; so $g_\alpha(\cdot, y)$ is a probability density for each $y \geqslant 0$.

Definition 3.6. For $\mu > 0$, $\alpha \in (0, \infty)^n$, and $Y = (Y_1, \ldots, Y_n)$ with $Y_i \geqslant 0$ a.s.,

$$h_{\alpha,\mu,Y}(x) := \mathsf{E} \prod_{i=1}^{n} \tfrac{1}{\mu}\, g_{\alpha_i}\big(\tfrac{x_i}{\mu}, Y_i\big), \qquad x \in (0, \infty)^n.$$

Lemma 3.7 (Properties of the h-family)**.** Write $h_\alpha := h_{\alpha,\mu,Y}$.

(i) $h_\alpha \geqslant 0$ and $\int h_\alpha \, \mathrm{d}x = 1$.

(ii) If $\alpha_i > 1$, then $h_\alpha(x) \to 0$ as $x_i \to 0+$, and

$$\frac{\partial}{\partial x_i} h_\alpha = h_{\alpha - e_i} - h_\alpha. \tag{10}$$

(iii) If $\alpha \in (1, \infty)^n$, then for any $J \subseteq [n]$ and $\lambda_i \geqslant 0$, $\partial^{|J|} h_\alpha / \partial x_J \in L_1((0, \infty)^n)$ and

$$\int e^{-\sum_i \lambda_i x_i} \frac{\partial^{|J|}}{\partial x_J} h_\alpha \, \mathrm{d}x \;=\; \Big(\prod_{i \in J} \lambda_i\Big) \int e^{-\sum_i \lambda_i x_i} h_\alpha \, \mathrm{d}x. \tag{11}$$

Proof.

(i) $\int_0^\infty \mu^{-1} g_\alpha(x_i/\mu, Y_i)\,\mathrm{d}x_i = \int_0^\infty g_\alpha(u, Y_i)\,\mathrm{d}u = 1$ for each ω (Theorem 3.5); Fubini gives $\int h_\alpha = \mathsf{E}[1] = 1$.

(ii) We split into two parts.

(a) Gamma bound. $\log\Gamma$ is strictly convex on $(0,\infty)$, and since $\Gamma(1) = \Gamma(2) = 1$, Γ has a unique minimum at some $s_0 \in (1,2)$ with $\Gamma(s_0) > \frac{1}{2}$ (numerically $s_0 \approx 1.462$, $\Gamma(s_0) \approx 0.886$). Hence

$$\Gamma(s) \geqslant \tfrac{1}{2} \quad \text{for all } s > 0, \tag{12}$$

giving $\Gamma(\alpha)^{-1} \leqslant 2$ for the $k = 0$ term. For $k \geqslant 1$, induction via $\Gamma(k+\alpha) = (k-1+\alpha)\Gamma(k-1+\alpha) \geqslant k \cdot \frac{1}{2}(k-2)! = \frac{1}{2}(k-1)!$ gives $\Gamma(k+\alpha) \geqslant \frac{1}{2}(k-1)!$ (base case $k = 1$: $\Gamma(1+\alpha) \geqslant \frac{1}{2}$). Substituting into $g_\alpha(x,y)$:

$$\begin{aligned} g_\alpha(x,y) &\leqslant e^{-x-y}\Big[2x^{\alpha-1} + \sum_{k\geqslant 1} \tfrac{2x^{k+\alpha-1}y^k}{(k-1)!k!}\Big] \\ &= 2x^{\alpha-1}e^{-x-y} + 2x^\alpha e^{-x} \underbrace{\sum_{k\geqslant 1} \tfrac{x^{k-1}}{(k-1)!} \cdot \tfrac{y^k e^{-y}}{k!}}_{\leqslant e^x \cdot 1} \leqslant 2x^{\alpha-1}(e^{-x} + x), \end{aligned}$$

using $y^k e^{-y}/k! \leqslant \sup_y(y^k e^{-y}/k!) \leqslant 1$ (term in a Poisson distribution), and $\sum_{k\geqslant 1} x^{k-1}/(k-1)! = e^x$. Thus

$$g_\alpha(x,y) \leqslant 2x^{\alpha-1}(e^{-x} + x), \tag{13}$$

and substituting into Theorem 3.6 (note Y_i drops out),

$$h_\alpha(x) \leqslant (2/\mu)^n \prod_i (x_i/\mu)^{\alpha_i - 1}(1 + x_i/\mu). \tag{14}$$

When $\alpha_i > 1$, the factor $(x_i/\mu)^{\alpha_i-1} \to 0$ as $x_i \to 0+$, so $h_\alpha(x) \to 0$.

(b) Derivative formula. Differentiating the series in Theorem 3.4 term by term (using $\Gamma(k+\alpha) = (k+\alpha-1)\Gamma(k+\alpha-1)$, valid when $\alpha > 0$),

$$\partial_x g_\alpha(x, y) = g_{\alpha-1}(x, y) - g_\alpha(x, y), \qquad \alpha > 1, \tag{15}$$

with convergence locally uniform by (13) applied to $g_{\alpha-1}$ (valid since $\alpha - 1 > 0$). Inserting into Theorem 3.6: the integrand and its x_i-derivative are bounded by an integrable function of (x, ω) uniformly in x_i on compact subsets of $(0, \infty)$ (use (13) for $g_{\alpha_i-1}(\cdot, Y_i)$ and $g_{\alpha_i}(\cdot, Y_i)$), justifying dominated convergence to differentiate under E, which gives (10).

(iii) Iterating (10),

$$\frac{\partial^{|J|}}{\partial x_J} h_\alpha = \sum_{\delta \in \{0,1\}^J} (-1)^{|J|-|\delta|} h_{\alpha - \sum_{i \in J} \delta_i e_i}, \tag{16}$$

each summand in L_1 by (i) (all $\alpha_i - \delta_i \geqslant \alpha_i - 1 > 0$). Integration by parts in each x_j, $j \in J$: the boundary term at $x_j \to \infty$ vanishes by (14) (exponential decay in every coordinate via $\int_0^\infty x^{\alpha-1}(1+x)e^{-ax}\,\mathrm{d}x < \infty$, implying pointwise decay of the partial integral); at $x_j \to 0+$ by (ii), applied to the $\alpha'_j := \alpha_j - \delta_j \geqslant$

$\alpha_j - 1 > 0$ shape, together with the extra factor $e^{-\sum_i \lambda_i x_i}$ which is bounded. Each integration by parts brings down a λ_j, giving (11).

□

3.5 The family $h_{k,C}$ and its Laplace transform

Definition 3.8. For $C \succ 0$ and $\mu > 0$ with $C - \mu I_n \succ 0$, set $A := (C - \mu I_n)^{1/2}$ (Theorem 1.3, which is well-defined and invertible), so that $C = \mu I_n + AA^\top$. Let $(g_j^{(l)})$, $j \leqslant n$, $l \leqslant k$, be i.i.d. $\mathcal{N}(0,1)$, and

$$Y_i := \sum_{l=1}^{k} \Big(\sum_{j=1}^{n} \tfrac{1}{\sqrt{2\mu}} g_j^{(l)} a_{ij} \Big)^2, \qquad h_{k,C} := h_{(k/2,\dots,k/2),\mu,Y}. \tag{17}$$

Lemma 3.9. For $\lambda_i \geqslant 0$, $\int e^{-\sum_i \lambda_i x_i} h_{k,C} \, \mathrm{d}x = |I_n + \Lambda C|^{-k/2}$.

Proof.

1. Scalar Laplace transform. Substitute $u := x/\mu$ and expand the series in Theorem 3.4:

$$\begin{aligned}\int_0^\infty \mu^{-1}e^{-\lambda x}g_\alpha(x/\mu, y)\,\mathrm{d}x &= \int_0^\infty e^{-\lambda\mu u-u-y}\sum_{k\geqslant 0}\tfrac{u^{k+\alpha-1}}{\Gamma(k+\alpha)}\tfrac{y^k}{k!}\,\mathrm{d}u \\ &= e^{-y}\sum_{k\geqslant 0}\tfrac{y^k}{k!}(1+\mu\lambda)^{-(k+\alpha)} \\ &= (1+\mu\lambda)^{-\alpha}\exp\bigl(-\tfrac{\mu\lambda}{1+\mu\lambda}y\bigr),\end{aligned}$$

using $\int_0^\infty u^{k+\alpha-1}e^{-(1+\mu\lambda)u}\,\mathrm{d}u = \Gamma(k+\alpha)(1+\mu\lambda)^{-(k+\alpha)}$ (Fubini-term-by-term justified by absolute convergence). So

$$\int_0^\infty \mu^{-1}e^{-\lambda x}g_\alpha(x/\mu, y)\,\mathrm{d}x = (1+\mu\lambda)^{-\alpha}\exp\bigl(-\tfrac{\mu\lambda}{1+\mu\lambda}y\bigr). \tag{18}$$

2. Product form. Applying (18) coordinatewise with $\alpha_i = k/2$ and Fubini (nonneg integrand),

$$\int e^{-\sum_i \lambda_i x_i}h_{k,C}\,\mathrm{d}x = \prod_i (1+\mu\lambda_i)^{-k/2}\cdot \mathsf{E}\exp\Bigl(-\sum_i \tfrac{\mu\lambda_i}{1+\mu\lambda_i}Y_i\Bigr). \tag{19}$$

3. Evaluate the expectation. From (17), set $X^{(l)} := \frac{1}{\sqrt{2\mu}}Ag^{(l)}$, so $X^{(l)} \sim \mathcal{N}(0, \frac{1}{2\mu}AA^\top)$ i.i.d. across l and

$Y_i = \sum_{l=1}^{k} (X_i^{(l)})^2$. By independence across l and Theorem 3.1 applied to each $X^{(l)}$ with $\tilde{\Lambda} := \operatorname{diag}(\frac{\mu\lambda_i}{1+\mu\lambda_i})$,

$$\mathsf{E}\exp\Big(-\sum_i \tfrac{\mu\lambda_i}{1+\mu\lambda_i} Y_i\Big) = \Big(\mathsf{E}\exp(-\langle \tilde{\Lambda} X^{(1)}, X^{(1)}\rangle)\Big)^k = |I_n + \tfrac{1}{\mu}\tilde{\Lambda} A A^\top|^{-k/2}.$$

4. Combine. Inserting step 3 into (19) and using $\prod_i (1+\mu\lambda_i)^{-k/2} = |I_n + \mu\Lambda|^{-k/2}$ (since $I_n + \mu\Lambda$ is diagonal),

$$\int e^{-\sum_i \lambda_i x_i} h_{k,C}\,\mathrm{d}x = \big(|I_n + \mu\Lambda| \cdot |I_n + \tfrac{1}{\mu}\tilde{\Lambda} A A^\top|\big)^{-k/2}.$$

The diagonal identity $\tilde{\Lambda}_i = \mu\lambda_i/(1+\mu\lambda_i)$ rearranges to $(I_n + \mu\Lambda)\tilde{\Lambda} = \mu\Lambda$. Hence, using the multiplicativity of det,

$$\begin{aligned}
|I_n + \mu\Lambda| \cdot |I_n + \tfrac{1}{\mu}\tilde{\Lambda} A A^\top| &= |(I_n + \mu\Lambda)(I_n + \tfrac{1}{\mu}\tilde{\Lambda} A A^\top)| \\
&= |I_n + \mu\Lambda + \tfrac{1}{\mu}(I_n + \mu\Lambda)\tilde{\Lambda}\, A A^\top| \\
&= |I_n + \mu\Lambda + \Lambda A A^\top| = |I_n + \Lambda(\mu I_n + A A^\top)| = |I_n + \Lambda C|,
\end{aligned}$$

giving the claim.

□

3.6 Uniqueness of the Laplace transform

Theorem 3.10. If $\varphi_1, \varphi_2 \in L_1((0,\infty)^n)$ satisfy $\int e^{-\sum_i \lambda_i x_i}\varphi_1 \,\mathrm{d}x = \int e^{-\sum_i \lambda_i x_i}\varphi_2 \,\mathrm{d}x$ for all $\lambda \in [0,\infty)^n$, then $\varphi_1 = \varphi_2$ a.e.

Proof. Set $\varphi := \varphi_1 - \varphi_2 \in L_1((0,\infty)^n)$; the hypothesis becomes $\int e^{-\sum_i \lambda_i x_i}\varphi \,\mathrm{d}x = 0$ for all $\lambda \in [0,\infty)^n$.

1. Change of variables. With $y_i := e^{-x_i}$, the map $(0,\infty)^n \to (0,1]^n$ has Jacobian $\prod_i y_i^{-1}$, so setting $\tilde{\varphi}(y) := \varphi(-\log y)\prod_i y_i^{-1} \in L_1((0,1]^n)$ and noting $e^{-\lambda_i x_i} = y_i^{\lambda_i}$,

$$\int_{(0,1]^n} y^\lambda \tilde{\varphi}(y)\,\mathrm{d}y = 0 \qquad \text{for all } \lambda \in [0,\infty)^n. \tag{20}$$

2. From polynomials to continuous functions. Restricting (20) to $\lambda \in \mathbb{Z}^n_{\geqslant 0}$ and taking finite linear combinations, $\int_{[0,1]^n} P(y)\tilde{\varphi}(y)\,\mathrm{d}y = 0$ for every polynomial P (extend $\tilde{\varphi}$ by zero at $y_i = 0$). By Stone–Weierstrass, polynomials are uniformly dense in $C([0,1]^n)$; given $\psi \in C([0,1]^n)$ and $\varepsilon > 0$, pick P with $\|\psi - P\|_\infty < \varepsilon$, so

$$\left|\int \psi\tilde{\varphi}\right| = \left|\int (\psi - P)\tilde{\varphi}\right| \leqslant \varepsilon\|\tilde{\varphi}\|_1.$$

   Sending $\varepsilon \to 0$: $\int_{[0,1]^n} \psi\tilde{\varphi}\,\mathrm{d}y = 0$ for every $\psi \in C([0,1]^n)$.
3. From continuous functions to $\tilde{\varphi} = 0$ a.e. Let $E := \{\tilde{\varphi} > 0\}$. By inner/outer regularity of Lebesgue measure, there exist compact $K_m \subseteq E$ and open $U_m \supseteq E$ with $|U_m \setminus K_m| \to 0$; Urysohn gives $\psi_m \in$

$C([0,1]^n)$ with $\mathbf{1}_{K_m} \leqslant \psi_m \leqslant \mathbf{1}_{U_m}$. Then $\psi_m \to \mathbf{1}_E$ a.e. along a subsequence, and $|\psi_m \tilde{\varphi}| \leqslant |\tilde{\varphi}| \in L_1$, so dominated convergence gives $\int_E \tilde{\varphi} = \lim \int \psi_m \tilde{\varphi} = 0$. Similarly for $\{\tilde{\varphi} < 0\}$, so $\int |\tilde{\varphi}| = 0$ and $\tilde{\varphi} = 0$ a.e., hence $\varphi = 0$ a.e.

□

4 Proof of Theorem 2.4

4.1 Reduction to monotonicity

With $s_i := \frac{1}{2}t_i^2$, $K := [0, s_1] \times \cdots \times [0, s_n]$, and $Z(\tau)$ as above, (2) is equivalent to $\mathsf{P}(Z(1) \in K) \geqslant \mathsf{P}(Z(0) \in K)$. At $\tau = 0$, $C(0) = \operatorname{diag}(C_{11}, C_{22})$ is block-diagonal, so the two blocks are independent and the RHS factorizes. At $\tau = 1$ we recover C. Hence Theorem 2.4 follows from:

Proposition 4.1. $\tau \mapsto \int_K f(x,\tau)\,\mathrm{d}x$ is non-decreasing on $[0,1]$.

4.2 Laplace-transform computation of $\partial_\tau f$

Theorem 4.1 amounts to $\int_K \partial_\tau f(x,\tau)\,\mathrm{d}x \geqslant 0$ for each $\tau \in (0,1)$ (Theorem 3.3(ii) with $\lambda = 0$ justifies $\partial_\tau \int_K f = \int_K \partial_\tau f$). Strategy:

(a) compute the Laplace transform of $\partial_\tau f$ and match it with that of a sum of derivatives of h_τ;

(b) use Laplace injectivity (Theorem 3.10) to get an a.e. pointwise identity;

(c) integrate this identity over K.

Step (a). Since $Z_i(\tau) = \frac{1}{2}X_i(\tau)^2$, applying Theorem 3.1 to $X(\tau) \sim \mathcal{N}(0, C(\tau))$ with $\lambda_i/2$ in place of λ_i,

$$\int_{[0,\infty)^n} e^{-\sum_i \lambda_i x_i} f(x,\tau)\,\mathrm{d}x = \mathsf{E}e^{-\sum_i \lambda_i Z_i(\tau)} = |I_n + \Lambda C(\tau)|^{-1/2}.$$

Differentiating in τ, with Theorem 3.3(ii) (taking $K = [0,\infty)^n$) justifying the interchange:

$$\int_{[0,\infty)^n} e^{-\sum_i \lambda_i x_i} \partial_\tau f(x,\tau)\,\mathrm{d}x = \partial_\tau |I_n + \Lambda C(\tau)|^{-1/2}. \tag{21}$$

Expand via Theorem 3.2(i) applied to $A = \Lambda C(\tau)$:

$$|I_n + \Lambda C(\tau)| = 1 + \sum_{\varnothing \neq J \subseteq [n]} |(\Lambda C(\tau))_J| = 1 + \sum_{\varnothing \neq J \subseteq [n]} |C(\tau)_J| \prod_{j \in J} \lambda_j, \tag{22}$$

using $|(\Lambda C(\tau))_J| = |\Lambda_J||C(\tau)_J| = \prod_{j\in J} \lambda_j \cdot |C(\tau)_J|$ since Λ is diagonal.

The block structure of $|C(\tau)_J|$. For $J \subseteq [n]$, set $J_1 := [n_1] \cap J$ and $J_2 := J \setminus [n_1]$. If $J_1 = \varnothing$ or $J_2 = \varnothing$, then $C(\tau)_J = C_J$ is τ-independent, since the off-diagonal blocks of $C(\tau)$ (where the τ appears) are not sampled. Otherwise, writing $C(\tau)_J$ in block form with blocks indexed by J_1, J_2,

$$C(\tau)_J = \begin{pmatrix} C_{J_1} & \tau C_{J_1 J_2} \\ \tau C_{J_2 J_1} & C_{J_2} \end{pmatrix},$$

and applying Theorem 3.2(ii),

$$|C(\tau)_J| = |C_{J_1}||C_{J_2}| \prod_{i=1}^{|J_1|} (1 - \tau^2 \mu_{J_1,J_2}(i)), \tag{23}$$

where the $\mu_{J_1,J_2}(i) \in [0,1]$ are the eigenvalues of $C_{J_1}^{-1/2} C_{J_1 J_2} C_{J_2}^{-1} C_{J_2 J_1} C_{J_1}^{-1/2}$ (bounded by Theorem 3.2(ii) since $C_J \succ 0$). Differentiating the product in τ,

$$\partial_\tau |C(\tau)_J| = -|C_{J_1}||C_{J_2}| \sum_{i=1}^{|J_1|} 2\tau \mu_{J_1,J_2}(i) \prod_{l \neq i} (1 - \tau^2 \mu_{J_1,J_2}(l)) \leqslant 0 \quad \text{on } [0,1].$$

Define

$$a_J(\tau) := -\partial_\tau |C(\tau)_J| \geqslant 0 \quad (\tau \in [0,1]), \tag{24}$$

with $a_J := 0$ if $J_1 = \varnothing$ or $J_2 = \varnothing$. Then

$$\partial_\tau |I_n + \Lambda C(\tau)| = -\sum_{\varnothing \neq J \subseteq [n]} a_J(\tau) \prod_{j \in J} \lambda_j. \tag{25}$$

Apply the chain rule $\partial_\tau (u^{-1/2}) = -\frac{1}{2} u^{-3/2}\, \partial_\tau u$ to $u = |I_n + \Lambda C(\tau)|$: (21) becomes

$$\int_{[0,\infty)^n} e^{-\sum_i \lambda_i x_i} \partial_\tau f(x,\tau)\, \mathrm{d}x = -\tfrac{1}{2} |I_n + \Lambda C(\tau)|^{-3/2} \partial_\tau |I_n + \Lambda C(\tau)|,$$

and substituting (25) (the two minus signs cancel):

$$\int_{[0,\infty)^n} e^{-\sum_i \lambda_i x_i} \partial_\tau f(x,\tau)\,\mathrm{d}x = \sum_{\varnothing \neq J \subseteq [n]} \tfrac{1}{2} a_J(\tau) |I_n + \Lambda C(\tau)|^{-3/2} \prod_{j \in J} \lambda_j. \tag{26}$$

4.3 Identification via $h_\tau := h_{3,C(\tau)}$

By Theorem 3.9 with $k = 3$ and Theorem 3.7(iii) (valid since $\alpha_i = \frac{3}{2} > 1$),

$$\int e^{-\sum_i \lambda_i x_i} \frac{\partial^{|J|}}{\partial x_J} h_\tau \,\mathrm{d}x = \Big(\prod_{j \in J} \lambda_j\Big) |I_n + \Lambda C(\tau)|^{-3/2}.$$

Comparing with (26): $\partial_\tau f(\cdot,\tau)$ and $\sum_J \frac{1}{2} a_J(\tau)\, \partial^{|J|} h_\tau / \partial x_J$ are L_1 with the same Laplace transform, so by Theorem 3.10,

$$\partial_\tau f(x,\tau) = \sum_{\varnothing \neq J \subseteq [n]} \tfrac{1}{2} a_J(\tau) \frac{\partial^{|J|}}{\partial x_J} h_\tau(x) \quad \text{a.e. on } (0,\infty)^n. \tag{27}$$

4.4 Integration over K

Integrate (27) over $K = \prod_j [0, s_j]$. For $J^c := [n] \setminus J$ and $(s_J, x_{J^c})_i = s_i$ $(i \in J)$, $= x_i$ $(i \in J^c)$, the fundamental theorem of calculus in each x_j, $j \in J$, gives

$$\int_K \frac{\partial^{|J|}}{\partial x_J} h_\tau(x)\,\mathrm{d}x = \int_{\prod_{j \in J^c}[0,s_j]} h_\tau(s_J, x_{J^c})\,\mathrm{d}x_{J^c} \geqslant 0, \tag{28}$$

since the lower-endpoint contributions vanish by Theorem 3.7(ii) ($\alpha_j = \frac{3}{2} > 1$), and $h_\tau \geqslant 0$. Since also $a_J \geqslant 0$,

$$\int_K \partial_\tau f(x,\tau)\,\mathrm{d}x = \sum_{\varnothing \neq J \subseteq [n]} \tfrac{1}{2} a_J(\tau) \int_{\prod_{j\in J^c}[0,s_j]} h_\tau(s_J, x_{J^c})\,\mathrm{d}x_{J^c} \geqslant 0,$$

which proves Theorem 4.1 and hence Theorem 2.4. □

5 Two corollaries

Corollary 5.1 (Šidák)**.** For $X \sim \mathcal{N}(0,C)$ and $s_i \geqslant 0$,

$$\mathsf{P}(|X_i| \leqslant s_i,\ i = 1,\dots,n) \geqslant \prod_{i=1}^n \mathsf{P}(|X_i| \leqslant s_i).$$

Proof. Induct on n. Case $n = 1$ is trivial. For the step, take $K_1 := \{|x_1| \leqslant s_1\}$ and $L_1 := \{|x_i| \leqslant s_i, 2 \leqslant i \leqslant n\}$ (both closed symmetric convex). Theorem 2.4 (applied with $n_1 = 1$, $n_2 = n-1$) gives $\mathsf{P}(X \in K_1 \cap L_1) \geqslant \mathsf{P}(X \in K_1)\mathsf{P}(X \in L_1)$; apply induction to $\mathsf{P}(X \in L_1)$. □

Corollary 5.2 (Decoupling)**.** For $X \sim \mathcal{N}(0,C)$ and closed symmetric convex $K, L \subseteq \mathbb{R}^n$,

$$\mathsf{P}(X \in K \cap L) \geqslant \mathsf{P}(X \in K)\,\mathsf{P}(X \in L).$$

Proof. Restatement of Theorem 1.1 for this K, L. $\square$